Free pythagorean theorem worksheet worksheet 16

Maths | 8th grade | Mixed Operations

1. If 3x + 5 = 2x - 4, what is the value of x?

A) 9
B) -9
C) 7
D) -7

2. Simplify: 4(3x - 2) + 5(x + 6)

A) 17x + 30
B) 7x + 28
C) 7x + 22
D) 12x + 28

3. Solve for y: 5y - 3(2y - 7) = 18

A) 4
B) 8
C) 10
D) 2

4. What is the value of 6^2 - 4 * 5?

A) 16
B) 36
C) 20
D) 8

5. Find the product of the roots of the equation: x^2 - 5x + 6 = 0.

A) 6
B) 5
C) -6
D) 1

6. What is the result of (3^3 - 5) / 2?

A) 11
B) 13
C) 7
D) 14

7. If a = 4 and b = 3, what is the value of 2a^2 - 3b?

A) 29
B) 41
C) 16
D) 35

8. Evaluate the expression: |7 - 12| + 5.

A) 0
B) 10

C) 8
D) 12

9. If 2(3w + 1) = 5w + 7, find the value of w.

A) 3
B) 5
C) 1
D) 2

10. The area of a rectangle is 48 square units. If the length is 8 units, what is the width?

A) 6
B) 8.5
C) 5
D) 9

Answer Key

1. D
Subtract 2x from both sides to get x + 5 = -4, then subtract 5 from both sides to solve for x = -9.

2. A
Distribute: 4(3x) - 4(2) + 5(x) + 5(6) = 12x - 8 + 5x + 30 = 17x + 22.

3. C
Distribute: 5y - 6y + 21 = 18. Combine like terms: -y + 21 = 18. Subtract 21: -y = -3. Divide by -1: y = 3.

4. A
Calculate: 6^2 = 36, 4 * 5 = 20, so 36 - 20 = 16.

5. A
The product of roots for ax^2 + bx + c = 0 is c/a. Here, c = 6 and a = 1, so 6/1 = 6.

6. C
Calculate: 3^3 = 27, 27 - 5 = 22, and 22 / 2 = 11.

7. D
Calculate: 2(4^2) - 3(3) = 2(16) - 9 = 32 - 9 = 23.

8. B
Calculate: |7 - 12| = |-5| = 5, then 5 + 5 = 10.

9. C
Distribute: 6w + 2 = 5w + 7. Subtract 5w: w + 2 = 7. Subtract 2: w = 5.

10. A
Area = length × width, 48 = 8 × width, so width = 48 / 8 = 6.